

SOLIDWORKS API (.NET VBA C#) — Reference (105/337): IFace2 (part 1 of 3)

SOLIDWORKS API Documentation (converted from CHM)

IFace2 (part 1 of 3) (Type)

SOLIDWORKS API Help

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IFace2 Interface

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[SolidWorks.Interop.sldworks Namespace](#) : IFace2 Interface

Allows access to the underlying edge, loop, and surface to the owning body or feature, and to face tessellation, trim data.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IFace2
```

Visual Basic (Usage)



```
Dim instance As IFace2
```

C#

```
public interface IFace2
```

C++/CLI

```
public interface class IFace2
```

Visual Basic for Applications (VBA) Syntax

See [Face2](#).

Example

[Get Faces Associated with Feature \(VBA\)](#)
[Traverse Bodies \(C++\)](#)
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Access Diagram

[Face2](#)

See Also

[IFace2 Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

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IFace2 Interface Methods

[See Also](#)

[SolidWorks.Interop.sldworks Namespace](#) : IFace2 Interface

For a list of all members of this type, see [IFace2 members](#).

Public Methods

Name	Description
 AddPropertyExtension	Adds a property extension to this face.
 AttachSurface	Attaches a surface to this face.
 CreateSheetBody	Creates a sheet body from this face.
 CreateSheetBodyByFaceExtension	Creates a sheet body by extending the face.
 DetachSurface	Detaches a surface from this face.
 EnumEdges	Enumerates the edges in a face.
 EnumLoops	Enumerates the loops in a face.
 FaceInSurfaceSense	Checks whether the face normal has the opposite direction (sense) as the underlying surface.
 GetAllAssemblyDecalProperties	Gets all of the decal properties applied to this face in an assembly component.
 GetAllDecalProperties	Gets all of the decal properties applied to this face in a part.
 GetArea	Gets the area of this face.
 GetBody	Gets the body containing this face.
 GetBox	Gets the box boundaries for this face.
 GetClosestPointOn	Uses the X,Y,Z input point to determine the closest point on the face.
 GetDecalsCount	Gets the number of decals applied to this face.
 GetEdgeCount	Gets the number of the edges that bound this face.
 GetEdges	Get the edges bounding this face.
 GetFacId	Gets the face ID of an imported body.
 GetFeature	Gets the feature that owns this face.
 GetFeatureId	Gets the order number for the owning feature of this face.
 GetFirstLoop	Gets the first loop in this face, which is not necessarily the outer loop.
 GetLoopCount	Gets the number of loops in this face.
 GetLoops	Gets all of the loops on this face.
 GetMaterialPropertyValues2	Gets the material property values for this face.
 GetNextFace	Gets the next face in a body.
 GetPatternSeedFeature	Gets the seed feature of a pattern.
 GetProjectedPointOn	Gets the point where the input point is projected on to this face.
 GetPropertyExtension	Gets the property extension on this face.
 GetSeedFeature	Gets the seed feature of a patterned, mirrored, or copied body for this face.
 GetShellType	Gets the shell type for this face.
 GetSilhouetteEdges	Obsolete. Superseded by IFace2.GetSilhouetteEdgesVB.
 GetSilhouetteEdgesVB	Gets the silhouette edges.

	GetSurface	Gets the surface referenced by this face.
	GetTessNorms	Gets the normal vector for each of the triangles that make up the shaded picture tessellation.
	GetTessTextures	Gets the texture coordinates for the triangles.
	GetTessTriangleCount	Gets the number of triangles that make up the shaded picture tessellation for this face.
	GetTessTriangles	Gets the triangles that make up the shaded picture tessellation for this face.
	GetTessTriStripEdges	Gets the edge ID for each of the edges in the triangles that make up the tessellation for this face.
	GetTessTriStripNorms	Gets the normal vector for each of the triangles that make up the shaded picture tessellation for this face.
	GetTessTriStrips	Gets the vertices that make up the shaded picture tessellation for this face.
	GetTessTriStripSize	Gets the array size required for IFace2::GetTessTriStrips and IFace2::IGetTessTriStrips .
	GetTessTriStripTextures	Gets the texture coordinate components for each vertex on each triangle strip on this face.
	GetTexture	Gets the texture applied to this face in the specified configuration.
	GetTrackingIDs	Gets the tracking IDs assigned to this face .
	GetTrackingIDsCount	Gets the number of tracking IDs on this face.
	GetTrimCurves	Obsolete. Superseded by IFace2::GetTrimCurves2
	GetTrimCurves2	Gets the definition of all of the entities that describe a trimmed face.
	GetTrimCurveTopology	Gets the trim curve topology for this face.
	GetTrimCurveTopologyCount	Gets the number of elements in the trim curve topology for this face.
	GetTrimCurveTopologyTypes	Gets the types of elements in the trim curve topology for this face.
	GetUVBounds	Gets the values that describe the U, V bounds of this face.
	HasMaterialPropertyValues	Gets whether this face has an appearance.
	HasTextureCoordinates	Gets whether this face has texture coordinates.
	Highlight	Adds highlighting to or removes highlighting from a face.
	ICreateSheetBody	Creates a sheet body from this face.
	ICreateSheetBodyByFaceExtension	Creates a sheet body by extending the face.
	IGetAllDecalProperties	Gets the decal properties applied to this face.
	IGetBody	Gets the body containing this face.
	IGetBox	Gets the box boundaries for this face.
	IGetClosestPointOn	Uses the X,Y,Z input point to determine the closest point on the face.
	IGetDecalProperties	Gets the properties of the specified decal on this face.
	IGetEdges	Get the edges bounding this face.
	IGetFaceData	Obsolete. Superseded by IFace2::GetTessNorms, IFace2::IGetTessNorms, IFace2::GetTessTextures, IFace2::IGetTessTextures, IFace2::GetTessTriangleCount, IFace2::GetTessTriangles, IFace2::IGetTessTriangles, IFace2::GetTessTriStripEdges, IFace2::IGetTessTriStripEdges, IFace2::GetTessTriStripNorms, IFace2::IGetTessTriStripNorms, IFace2::GetTessTriStrips, and IFace2::IGetTessTriStrips
	IGetFeature	Gets the feature that owns this face.
	IGetFirstLoop	Gets the first loop in this face, which is not necessarily the outer loop.
	IGetLoops	Gets all of the loops on this face.
	IGetMaterialPropertyValues2	Gets the material property values for this face.
	IGetNextFace	Gets the next face in a body.
	IGetPatternSeedFeature	Gets the seed feature of a pattern.
	IGetSilhouetteEdgeCount	Gets the number of silhouette edges for this face.
	IGetSilhouetteEdges	Gets the silhouette edges for this face with the specified root point and in the specified direction.
	IGetSurface	Gets the surface referenced by this face.
	IGetTessNorms	Gets the normal vector for each of the triangles that make up the shaded picture tessellation.
	IGetTessTextures	Gets the texture coordinates for the triangles.
	IGetTessTriangles	Gets the triangles that make up the shaded picture tessellation for this face.

		
	IGetTessTriStripEdges	Gets the edge ID for each of the edges in the triangles that make up the tessellation for this face.
	IGetTessTriStripEdgeSize	Gets the size of the array returned by IFace2::GetTessTriStripEdges and IFace2::IGetTessTriStripEdges .
	IGetTessTriStripNorms	Gets the normal vector for each of the triangles that make up the shaded picture tessellation for this face.
	IGetTessTriStrips	Gets the vertices that make up the shaded picture tessellation for this face.
	IGetTrackingIDs	Gets the tracking IDs assigned to this face .
	IGetTrimCurveSize	Obsolete. Superseded by IFace2::IGetTrimCurveSize2
	IGetTrimCurveSize2	Gets the size of the array required for IFace2::GetTrimCurves2
	IGetTrimCurveTopology	Gets the trim curve topology for this face.
	IGetTrimCurveTopologyTypes	Gets the trim curve topology type array for this face.
	IGetUVBounds	Gets the values that describe the U, V bounds of this face.
	IHighlight	Adds highlighting to or removes highlighting from a face.
	IImprintCurve	Imprints a curve on the selected face.
	IsCoincident	Gets whether this face and the specified face, which is located on a different body, are coincident.
	IsSame	Gets whether the two faces are the same.
	ImprintCurve	Imprints a curve on the selected face.
	ImprintCurveCount	Gets the number of new edges and faces to create when imprinting a curve.
	IRemoveInnerLoops	Removes the inner loops on this face if the face is from a sheet (surface) body.
	IRemoveMaterialProperty2	Removes the material property values from this face.
	IReverseEvaluate	Gets the UV parameters for the given XYZ location on this face.
	IsCoincident	Gets whether this face and the specified face, which is located on a different body, are coincident.
	ISetMaterialPropertyValues2	Sets the material property values for this face.
	IsSame	Gets whether the two faces are the same.
	RemoveFaceId	Removes the face ID on an imported body.
	RemoveInnerLoops	Removes the inner loops on this face if the face is from a sheet (surface) body.
	RemoveMaterialProperty	Obsolete. Superseded by IFace2::RemoveMaterialProperty2
	RemoveMaterialProperty2	Removes the material property values from this face.
	RemoveRedundantTopology	Removes redundant topology from the face.
	RemoveTexture	Obsolete. Superseded by IFace2::RemoveTexture2
	RemoveTexture2	Removes the texture applied to this face in the specified configuration.
	RemoveTextureByDisplayState	Removes the texture applied to this face in the specified display state.
	RemoveTrackingID	Removes a tracking ID assigned to this face .
	ResetPropertyExtension	Clears all values stored in the property extension.
	ReverseEvaluate	Gets the UV parameters for the given XYZ location on this face.
	SetFaceId	Sets the face ID on an imported body.
	SetMaterialPropertyValues2	Sets the material property values for this face.
	SetTexture	Applies texture to this face in the specified configuration.
	SetTextureByDisplayState	Applies texture to this face in the specified display state.
	SetTrackingID	Assigns a tracking ID to this face.

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IFace2 Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : IFace2 Interface

The following tables list the members exposed by [IFace2](#).

Public Properties

Name	Description
Check	Checks whether the face is a valid, and, if not, returns the faults.
IMaterialPropertyValues	Gets or sets the material properties for the selected face in the active configuration.
INormal	Gets the unit normal vector for any planar face. NOTE: This property is a get-only property. Set is not implemented.
MaterialIdName	Gets or sets the material name.
MaterialPropertyValues	Gets or sets the material properties for the selected face in the active configuration.
MaterialUserName	Gets or sets the material name, which is visible to the user.
Normal	Gets the unit normal vector for any planar face. NOTE: This property is a get-only property. Set is not implemented.

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Public Methods

Name	Description
AddPropertyExtension	Adds a property extension to this face.
AttachSurface	Attaches a surface to this face.
CreateSheetBody	Creates a sheet body from this face.
CreateSheetBodyByFaceExtension	Creates a sheet body by extending the face.
DetachSurface	Detaches a surface from this face.
EnumEdges	Enumerates the edges in a face.
EnumLoops	Enumerates the loops in a face.
FaceInSurfaceSense	Checks whether the face normal has the opposite direction (sense) as the underlying surface.
GetAllAssemblyDecalProperties	Gets all of the decal properties applied to this face in an assembly component.
GetAllDecalProperties	Gets all of the decal properties applied to this face in a part.
GetArea	Gets the area of this face.
GetBody	Gets the body containing this face.
GetBox	Gets the box boundaries for this face.
GetClosestPointOn	Uses the X,Y,Z input point to determine the closest point on the face.
GetDecalsCount	Gets the number of decals applied to this face.
GetEdgeCount	Gets the number of the edges that bound this face.
GetEdges	Get the edges bounding this face.
GetFacetId	Gets the face ID of an imported body.
GetFeature	Gets the feature that owns this face.
GetFeatureId	Gets the order number for the owning feature of this face.
GetFirstLoop	Gets the first loop in this face, which is not necessarily the outer loop.
GetLoopCount	Gets the number of loops in this face.

		
	GetLoops	Gets all of the loops on this face.
	GetMaterialPropertyValues2	Gets the material property values for this face.
	GetNextFace	Gets the next face in a body.
	GetPatternSeedFeature	Gets the seed feature of a pattern.
	GetProjectedPointOn	Gets the point where the input point is projected on to this face.
	GetPropertyExtension	Gets the property extension on this face.
	GetSeedFeature	Gets the seed feature of a patterned, mirrored, or copied body for this face.
	GetShellType	Gets the shell type for this face.
	GetSilhoutteEdges	Obsolete. Superseded by IFace2::GetSilhoutteEdgesVB .
	GetSilhoutteEdgesVB	Gets the silhouette edges.
	GetSurface	Gets the surface referenced by this face.
	GetTessNorms	Gets the normal vector for each of the triangles that make up the shaded picture tessellation.
	GetTessTextures	Gets the texture coordinates for the triangles.
	GetTessTriangleCount	Gets the number of triangles that make up the shaded picture tessellation for this face.
	GetTessTriangles	Gets the triangles that make up the shaded picture tessellation for this face.
	GetTessTriStripEdges	Gets the edge ID for each of the edges in the triangles that make up the tessellation for this face.
	GetTessTriStripNorms	Gets the normal vector for each of the triangles that make up the shaded picture tessellation for this face.
	GetTessTriStrips	Gets the vertices that make up the shaded picture tessellation for this face.
	GetTessTriStripSize	Gets the array size required for IFace2::GetTessTriStrips and IFace2::IGetTessTriStrips .
	GetTessTriStripTextures	Gets the texture coordinate components for each vertex on each triangle strip on this face.
	GetTexture	Gets the texture applied to this face in the specified configuration.
	GetTrackingIDs	Gets the tracking IDs assigned to this face .
	GetTrackingIDsCount	Gets the number of tracking IDs on this face.
	GetTrimCurves	Obsolete. Superseded by IFace2::GetTrimCurves2 .
	GetTrimCurves2	Gets the definition of all of the entities that describe a trimmed face.
	GetTrimCurveTopology	Gets the trim curve topology for this face.
	GetTrimCurveTopologyCount	Gets the number of elements in the trim curve topology for this face.
	GetTrimCurveTopologyTypes	Gets the types of elements in the trim curve topology for this face.
	GetUVBounds	Gets the values that describe the U, V bounds of this face.
	HasMaterialPropertyValues	Gets whether this face has an appearance.
	HasTextureCoordinates	Gets whether this face has texture coordinates.
	Highlight	Adds highlighting to or removes highlighting from a face.
	ICreateSheetBody	Creates a sheet body from this face.
	ICreateSheetBodyByFaceExtension	Creates a sheet body by extending the face.
	IGetAllDecalProperties	Gets the decal properties applied to this face.
	IGetBody	Gets the body containing this face.
	IGetBox	Gets the box boundaries for this face.
	IGetClosestPointOn	Uses the X,Y,Z input point to determine the closest point on the face.
	IGetDecalProperties	Gets the properties of the specified decal on this face.
	IGetEdges	Get the edges bounding this face.
	IGetFacetData	Obsolete. Superseded by IFace2::GetTessNorms , IFace2::IGetTessNorms , IFace2::GetTessTextures , IFace2::IGetTessTextures , IFace2::GetTessTriangleCount , IFace2::GetTessTriangles , IFace2::IGetTessTriangles , IFace2::GetTessTriStripEdges , IFace2::IGetTessTriStripEdges , IFace2::GetTessTriStripNorms , IFace2::IGetTessTriStripNorms , IFace2::GetTessTriStrips , and IFace2::IGetTessTriStrips .
	IGetFeature	Gets the feature that owns this face.

		
	IGetFirstLoop	Gets the first loop in this face, which is not necessarily the outer loop.
	IGetLoops	Gets all of the loops on this face.
	IGetMaterialPropertyValues2	Gets the material property values for this face.
	IGetNextFace	Gets the next face in a body.
	IGetPatternSeedFeature	Gets the seed feature of a pattern.
	IGetSilhouetteEdgeCount	Gets the number of silhouette edges for this face.
	IGetSilhouetteEdges	Gets the silhouette edges for this face with the specified root point and in the specified direction.
	IGetSurface	Gets the surface referenced by this face.
	IGetTessNorms	Gets the normal vector for each of the triangles that make up the shaded picture tessellation.
	IGetTessTextures	Gets the texture coordinates for the triangles.
	IGetTessTriangles	Gets the triangles that make up the shaded picture tessellation for this face.
	IGetTessTriStripEdges	Gets the edge ID for each of the edges in the triangles that make up the tessellation for this face.
	IGetTessTriStripEdgeSize	Gets the size of the array returned by IFace2::GetTessTriStripEdges and IFace2::IGetTessTriStripEdges .
	IGetTessTriStripNorms	Gets the normal vector for each of the triangles that make up the shaded picture tessellation for this face.
	IGetTessTriStrips	Gets the vertices that make up the shaded picture tessellation for this face.
	IGetTrackingIDs	Gets the tracking IDs assigned to this face .
	IGetTrimCurveSize	Obsolete. Superseded by IFace2::IGetTrimCurveSize2
	IGetTrimCurveSize2	Gets the size of the array required for IFace2::GetTrimCurves2
	IGetTrimCurveTopology	Gets the trim curve topology for this face.
	IGetTrimCurveTopologyTypes	Gets the trim curve topology type array for this face.
	IGetUVBounds	Gets the values that describe the U, V bounds of this face.
	IHighlight	Adds highlighting to or removes highlighting from a face.
	IImprintCurve	Imprints a curve on the selected face.
	IsCoincident	Gets whether this face and the specified face, which is located on a different body, are coincident.
	IsSame	Gets whether the two faces are the same.
	ImprintCurve	Imprints a curve on the selected face.
	ImprintCurveCount	Gets the number of new edges and faces to create when imprinting a curve.
	IRemoveInnerLoops	Removes the inner loops on this face if the face is from a sheet (surface) body.
	IRemoveMaterialProperty2	Removes the material property values from this face.
	IReverseEvaluate	Gets the UV parameters for the given XYZ location on this face.
	IsCoincident	Gets whether this face and the specified face, which is located on a different body, are coincident.
	ISetMaterialPropertyValues2	Sets the material property values for this face.
	IsSame	Gets whether the two faces are the same.
	RemoveFaceId	Removes the face ID on an imported body.
	RemoveInnerLoops	Removes the inner loops on this face if the face is from a sheet (surface) body.
	RemoveMaterialProperty	Obsolete. Superseded by IFace2::RemoveMaterialProperty2
	RemoveMaterialProperty2	Removes the material property values from this face.
	RemoveRedundantTopology	Removes redundant topology from the face.
	RemoveTexture	Obsolete. Superseded by IFace2::RemoveTexture2
	RemoveTexture2	Removes the texture applied to this face in the specified configuration.
	RemoveTextureByDisplayState	Removes the texture applied to this face in the specified display state.

	RemoveTrackingID	Removes a tracking ID assigned to this face .
	ResetPropertyExtension	Clears all values stored in the property extension.
	ReverseEvaluate	Gets the UV parameters for the given XYZ location on this face.
	SetFaceId	Sets the face ID on an imported body.
	SetMaterialPropertyValues2	Sets the material property values for this face.
	SetTexture	Applies texture to this face in the specified configuration.
	SetTextureByDisplayState	Applies texture to this face in the specified display state.
	SetTrackingID	Assigns a tracking ID to this face.

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See Also

[IFace2 Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

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IFace2 Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IFace2 Interface

For a list of all members of this type, see [IFace2 members](#).

Public Properties

Name	Description
 Check	Checks whether the face is a valid, and, if not, returns the faults.
 IMaterialPropertyValues	Gets or sets the material properties for the selected face in the active configuration.
 INormal	Gets the unit normal vector for any planar face. NOTE: This property is a get-only property. Set is not implemented.
 MaterialIdName	Gets or sets the material name.
 MaterialPropertyValues	Gets or sets the material properties for the selected face in the active configuration.
 MaterialUserName	Gets or sets the material name, which is visible to the user.
 Normal	Gets the unit normal vector for any planar face. NOTE: This property is a get-only property. Set is not implemented.

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See Also

[IFace2 Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

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AddPropertyExtension Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : AddPropertyExtension Method (IFace2)

Adds a property extension to this face.

.NET Syntax

Visual Basic (Declaration)

```
Function AddPropertyExtension( _
    ByVal PropertyExtension As System.Object _
) As System.Integer
```

```
Dim instance As IFace2
Dim PropertyExtension As System.Object
Dim value As System.Integer

value = instance.AddPropertyExtension(PropertyExtension)
```

C#

```
System.int AddPropertyExtension(
    System.object PropertyExtension
)
```

C++/CLI

```
System.int AddPropertyExtension(
    & System.Object^ PropertyExtension
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PropertyExtension
Value of the property extension to add to this face (see **Remarks**)

Return Value

1 if the property extension is added to the face, -1 if not

Visual Basic for Applications (VBA) Syntax

See [Face2::AddPropertyExtension](#).

Remarks

This method does not support:

- adding multiple property extensions to the same face.
- faces obtained from reference surface bodies.

To use this method:

1. Declare the variable.
2. Assign the variable a value: float, integer, or string.
3. Call this method to add the value to the face.

NOTE: SOLIDWORKS recommends that you use the [IAttribute](#), [IAttributeDef](#), and [IParameter](#) interfaces instead of this method. These interface provide more flexibility.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetPropertyExtension Method](#)
[IFace2::ResetPropertyExtension Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

AttachSurface Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : AttachSurface Method (IFace2)

Attaches a surface to this face.

.NET Syntax

```
Function AttachSurface( _
    ByVal SurfIn As Surface, _
    ByVal SenseIn As System.Boolean _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IFace2
Dim SurfIn As Surface
Dim SenseIn As System.Boolean
Dim value As System.Boolean

value = instance.AttachSurface(SurfIn, SenseIn)
```

C#

```
System.bool AttachSurface(
    Surface SurfIn,
    System.bool SenseIn
)
```

C++/CLI

```
System.bool AttachSurface(
    & Surface^ SurfIn,
    & System.bool SenseIn
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SurfIn

[Surface](#) to attach to face

SenseIn

True if surface normal and face normal are in the same direction, false if surface normal and face normal are in opposite direction

Return Value

True if surface is attached to face, false if not

Visual Basic for Applications (VBA) Syntax

See [Face2::AttachSurface](#).

Remarks

To detach surfaces from faces, use [IFace2::DetachSurface](#). Neither IFace2::AttachSurface nor IFace2::DetachSurface affect body topology.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetSurface Method](#)
[IFace2::IGetSurface Method](#)
[IFace2::HasTextureCoordinates Method \(\)](#)

Availability

SOLIDWORKS 2004 SP4, Revision Number 12.4

SOLIDWORKS API Help

[Send comments on this topic.](#)

Check Property (IFace2)

See Also [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : Check Property (IFace2)

Checks whether the face is a valid, and, if not, returns the faults.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property Check As FaultEntity

Visual Basic (Usage)



```
Dim instance As IFace2
Dim value As FaultEntity

value = instance.Check
```

C#

```
FaultEntity Check {get;}
```

C++/CLI

```
property FaultEntity^ Check {
    FaultEntity^ get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Pointer to [IFaultEntity](#) object

Visual Basic for Applications (VBA) Syntax

See [Face2::Check](#).

Example

[Check Faces for Faults \(VBA\)](#)

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IEdge::Check Property](#)
[IBody2::Check3 Property](#)

Availability

SOLIDWORKS 2004 SP4, Revision Number 12.4

SOLIDWORKS API Help

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CreateSheetBody Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : CreateSheetBody Method (IFace2)

Creates a sheet body from this face.

.NET Syntax

Visual Basic (Declaration)

Function CreateSheetBody() As System.Object

Visual Basic (Usage)



```
Dim instance As IFace2
Dim value As System.Object

value = instance.CreateSheetBody()
```

C#

```
System.object CreateSheetBody()
```

```
System.Object^ CreateSheetBody();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

New sheet [body](#).

Visual Basic for Applications (VBA) Syntax

See [Face2::CreateSheetBody](#).

Remarks

A sheet body is a surface body.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::ICreateSheetBody Method](#)
[IFace2::ICreateSheetBodyByFaceExtension Method](#)
[IFace2::CreateSheetBodyByFaceExtension Method](#)
[IModeler::CreateSheetFromFaces Method](#)
[IModeler::ICreateSheetFromFaces Method](#)
[IBody2::DeleteFacesMakeSheetBodies Method](#)
[IBody2::IDeleteFacesMakeSheetBodies Method](#)
[IBody2::IDeleteFacesMakeSheetBodiesCount Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateSheetBodyByFaceExtension Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : CreateSheetBodyByFaceExtension Method (IFace2)

Creates a sheet body by extending the face.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateSheetBodyByFaceExtension( _
    ByVal BoxLowIn As System.Object, _
    ByVal BoxHighIn As System.Object _
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As IFace2
Dim BoxLowIn As System.Object
Dim BoxHighIn As System.Object
Dim value As System.Object

value = instance.CreateSheetBodyByFaceExtension(BoxLowIn, BoxHighIn)
```

C#

```
System.object CreateSheetBodyByFaceExtension(
    System.object BoxLowIn,
    System.object BoxHighIn
)
```

```
System.Object^ CreateSheetBodyByFaceExtension(  
& System.Object^ BoxLowIn,  
& System.Object^ BoxHighIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BoxLowIn

Array of 3 doubles (x,y,z)

BoxHighIn

Array of 3 doubles (x,y,z)

Return Value

Newly created sheet body

Visual Basic for Applications (VBA) Syntax

See [Face2::CreateSheetBodyByFaceExtension](#).

Remarks

A sheet body is a surface body.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::CreateSheetBodyByFaceExtension Method](#)
[IFace2::CreateSheetBody Method](#)
[IFace2::CreateSheetBody Method](#)
[IBody2::DeleteFacesMakeSheetBodies Method](#)
[IBody2::DeleteFacesMakeSheetBodies Method](#)
[IModeler::CreateSheetFromFaces Method](#)
[IModeler::CreateSheetFromFaces Method](#)
[IBody2::DeleteFacesMakeSheetBodiesCount Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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DetachSurface Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : DetachSurface Method (IFace2)

Detaches a surface from this face.

.NET Syntax

Visual Basic (Declaration)

```
Function DetachSurface() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As System.Boolean  
  
value = instance.DetachSurface()
```

C#

```
System.bool DetachSurface()
```

C++/CLI

```
System.bool DetachSurface();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the surface is detached from this face, false if not

Visual Basic for Applications (VBA) Syntax

See [Face2::DetachSurface](#).

Example

Use [Face2::AttachSurface](#) to attach surfaces to faces. Neither `IFace2::DetachFace` nor `IFace2::AttachFace` affect body topology.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::IGetSurface Method](#)
[IFace2::GetSurface Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

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EnumEdges Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : EnumEdges Method (IFace2)

Enumerates the edges in a face.

.NET Syntax

Visual Basic (Declaration)

```
Function EnumEdges() As EnumEdges
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As EnumEdges  
  
value = instance.EnumEdges()
```

C#

```
EnumEdges EnumEdges()
```

C++/CLI

```
EnumEdges^ EnumEdges();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[Edges enumeration](#)

Visual Basic for Applications (VBA) Syntax

See [Face2::EnumEdges](#).

Remarks

If you want to align the edges with their corresponding trim curves, then call this method immediately following a call to [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#). This causes the order of the trim curves to align with the order of the edges. For Dispatch implementations, use [IFace2::GetEdges](#).

To elaborate, any call to [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#) stores a temporary buffer containing the edges for this face. If this temporary buffer exists, then [IFace2::GetEdges](#), [IFace2::IGetEdges](#), or [IFace2::EnumEdges](#) access the buffer. This is useful when you want to align the trim curves of the face with their corresponding edges. If [IFace2::GetEdges](#), [IFace2::IGetEdges](#), or [IFace2::EnumEdges](#) is called immediately after [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#), then the order of the trim curves align with the order of the edges. However, if you are processing all the trim curves and then processing all the edges, your first call to [IFace2::EnumEdges](#) uses the temporary buffer of edges generated by your last call to [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#). Because your last call to [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#) and your current call to [IFace2::EnumEdges](#) are probably made from different faces, the data does not match. You can see this mismatch of data when [IFace2::GetEdgeCount](#) does not match the number of enumerated edges, or when the edge data does not match the examined face. Therefore, call [IFace2::GetEdges](#), [IFace2::IGetEdges](#), or [IFace2::EnumEdges](#) in the same loop and immediately after your call to [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#). The temporary buffer of edges is destroyed after each call to [IFace2::GetEdges](#), [IFace2::IGetEdges](#), or [IFace2::EnumEdges](#).

If you are using [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#), the number of trim curves on a face might vary from the number of edges on the face. If you call [IFace2::EnumEdges](#) immediately after calling [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#), then you are getting list of edges generated by [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#). Because this list of generated edges might vary from the list of edges that are typically returned from this face, do not use [IFace2::GetEdgeCount](#) to determine the length of the enumerated edge list. Instead, use one of the return values from [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#) to determine the size of the enumerated edge list. [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#) returns the number of trim (SP) curves in the first array position. This is the value you can use to loop through your enumerated list of edges.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetEdges Method](#)
[IFace2::IGetEdges Method](#)
[IFace2::GetEdgeCount Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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EnumLoops Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : EnumLoops Method (IFace2)

Enumerates the loops in a face.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function EnumLoops() As EnumLoops2
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As EnumLoops2
```

```
value = instance.EnumLoops()
```

C#

```
EnumLoops2 EnumLoops()
```

C++/CLI

```
EnumLoops2^ EnumLoops();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the [loops enumeration](#)

Visual Basic for Applications (VBA) Syntax

See [Face2::EnumLoops](#).

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetFirstLoop Method](#)
[IFace2::GetLoopCount Method](#)
[IFace2::!GetFirstLoop Method](#)
[IFace2::RemoveInnerLoops Method](#)
[IFace2::!RemoveInnerLoops Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

FaceInSurfaceSense Method (IFace2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : FaceInSurfaceSense Method (IFace2)

Checks whether the face normal has the opposite direction (sense) as the underlying surface.

.NET Syntax

Visual Basic (Declaration)

```
Function FaceInSurfaceSense() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As System.Boolean  
  
value = instance.FaceInSurfaceSense()
```

C#

```
System.bool FaceInSurfaceSense()
```

C++/CLI

```
System.bool FaceInSurfaceSense();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if face normal and surface normal are in the opposite direction, false if they are in same direction

Visual Basic for Applications (VBA) Syntax

See [Face2::FaceInSurfaceSense](#).

Example

[Determine Type of Face \(VBA\)](#)
[Get Normal of Planar Surface \(VBA\)](#)
[Select Edges of All Holes on Face \(C#\)](#)
[Select Edges of All Holes on Face \(VB.NET\)](#)
[Select Edges of All Holes on Face \(VBA\)](#)

Remarks

Although the name of this method and its results are somewhat contradictory, the results are:

- True if face normal and surface normal are in the opposite direction.
- False if they are in the same direction.

You might need this method because the underlying surface geometry can have an orientation where its normal vector points toward or away from the body material. The normal vector of faces, however, are directed away from the body material.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetAllAssemblyDecalProperties Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetAllAssemblyDecalProperties Method (IFace2)

Gets all of the decal properties applied to this face in an assembly component.

.NET Syntax

Visual Basic (Declaration)

```
Function GetAllAssemblyDecalProperties() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As System.Object  
  
value = instance.GetAllAssemblyDecalProperties()
```

C#

```
System.object GetAllAssemblyDecalProperties()
```

C++/CLI

```
System.Object^ GetAllAssemblyDecalProperties();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [IFaceDecalProperties](#) objects

Visual Basic for Applications (VBA) Syntax

See [Face2::GetAllAssemblyDecalProperties](#).

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetAllDecalProperties Method \(\)](#)

Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetAllDecalProperties Method (IFace2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetAllDecalProperties Method (IFace2)

Gets all of the decal properties applied to this face in a part.

.NET Syntax

Visual Basic (Declaration)

```
Function GetAllDecalProperties() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As System.Object  
  
value = instance.GetAllDecalProperties()
```

C#

```
System.object GetAllDecalProperties()
```

C++/CLI

```
System.Object^ GetAllDecalProperties();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [decal properties applied to this face](#)

Visual Basic for Applications (VBA) Syntax

See [Face2::GetAllDecalProperties](#).

Example

[Add Decal \(C#\)](#)
[Add Decal \(VB.NET\)](#)
[Add Decal \(VBA\)](#)

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetDecalsCount Method](#)
[IFace2::GetAllAssemblyDecalProperties Method \(\)](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetArea Method (IFace2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetArea Method (IFace2)

Gets the area of this face.

.NET Syntax

Visual Basic (Declaration)

```
Function GetArea() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As System.Double  
  
value = instance.GetArea()
```

C#

```
System.double GetArea()
```

C++/CLI

```
System.double GetArea();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Face area in square meters (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Face2::GetArea](#).

Example

[Find Outside Edges of Face \(VBA\)](#)
[Sweep Planar Loop Along Vector \(VBA\)](#)
[Get Areas of MidSurface Faces \(C#\)](#)
[Get Areas of MidSurface Faces \(VB.NET\)](#)
[Get Areas of MidSurface Faces \(VBA\)](#)

Remarks

This method calculates and returns an approximate value. If your application requires a more accurate value for a face area, then use tessellation. See [IBody2::GetTessellation](#) and [ITessellation](#) for details.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetBody Method (IFace2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetBody Method (IFace2)

Gets the body containing this face.

.NET Syntax

Visual Basic (Declaration)

```
Function GetBody() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As System.Object  
  
value = instance.GetBody()
```

C#

```
System.object GetBody()
```

C++/CLI

```
System.Object^ GetBody();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[Body](#)

Visual Basic for Applications (VBA) Syntax

See [Face2::GetBody](#).

Example

[Create and Modify Dome Feature \(C#\)](#)
[Create and Modify Dome Feature \(VB.NET\)](#)
[Create and Modify Dome Feature \(VBA\)](#)

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::IGetBody Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetBox Method (IFace2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetBox Method (IFace2)

Gets the box boundaries for this face.

.NET Syntax

Visual Basic (Declaration)

```
Function GetBox() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As System.Object  
  
value = instance.GetBox()
```

C#

```
System.object GetBox()
```

C++/CLI

```
System.Object^ GetBox();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Object containing the two diagonal points that bound the component; an array of 6 doubles

Visual Basic for Applications (VBA) Syntax

See [Face2::GetBox](#).

Example

[Get Bounding Box \(C#\)](#)
[Get Bounding Box \(VB.NET\)](#)
[Get Bounding Box \(VBA\)](#)

Remarks

IMPORTANT: The values returned are approximate and should not be used for comparison or calculation purposes. Furthermore, the bounding box may vary after rebuilding the model.

The resulting box encloses the object, but it might not be the tightest box.

This method returns X,Y,Z points that are the lower and upper diagonal corners that bound the face with the box sides parallel to the X, Y and Z axes. The box dimensions enclose the face and are typically close to the minimum possible size, but this is not guaranteed.

The return value is an array of doubles as follows:

[XCorner1, YCorner1, ZCorner1, XCorner2, YCorner2, ZCorner2]

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::IGetBox Method](#)
[IFeature::GetBox Method](#)
[IFeature::IGetBox Method](#)
[IAssemblyDoc::GetBox Method](#)
[IAssemblyDoc::IGetBox Method](#)
[IComponent2::GetBox Method](#)
[IComponent2::IGetBox Method](#)
[IBody2::GetBodyBox Method](#)
[IBody2::IGetBodyBox Method](#)
[IModelDocExtension::GetVisibleBox Method](#)
[IModelDocExtension::RemoveVisibleBox Method](#)
[IModelDocExtension::SetVisibleBox Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetClosestPointOn Method (IFace2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetClosestPointOn Method (IFace2)

Uses the X,Y,Z input point to determine the closest point on the face.

.NET Syntax

Visual Basic (Declaration)

```
Function GetClosestPointOn( _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As System.Object  
  
value = instance.GetClosestPointOn(X, Y, Z)
```

C#

```
System.object GetClosestPointOn(  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
System.Object^ GetClosestPointOn(  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X
X value of the input point

Y
Y value of the input point

Z
Z value of the input point

Return Value

Array of five doubles representing the X,Y,Z point on the face followed by the U, V parameter on the face that is closest to the input point

Visual Basic for Applications (VBA) Syntax

See [Face2::GetClosestPointOn](#).

Example

[Calculate Closest Distance Between Faces \(VBA\)](#)

Remarks

This method returns only one point, regardless of how many points achieve the minimum distance.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::IGetClosestPointOn Method](#)
[ICurve::GetClosestPointOn Method](#)
[ICurve::IGetClosestPointOn Method](#)
[IEdge::GetClosestPointOn Method](#)
[IEdge::IGetClosestPointOn Method](#)
[IFace2::GetProjectedPointOn Method](#)
[ISurface::GetClosestPointOn Method](#)
[ISurface::IGetClosestPointOn Method](#)
[IVertex::GetClosestPointOn Method](#)
[IVertex::IGetClosestPointOn Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetDecalsCount Method (IFace2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetDecalsCount Method (IFace2)

Gets the number of decals applied to this face.

.NET Syntax

Visual Basic (Declaration)

```
Function GetDecalsCount() As System.Integer
```

Visual Basic (Usage)

```
Dim instance As IFace2  
Dim value As System.Integer  
  
value = instance.GetDecalsCount()
```

C#

```
System.int GetDecalsCount()
```

C++/CLI

```
System.int GetDecalsCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of decals applied to this face

Visual Basic for Applications (VBA) Syntax

See [Face2::GetDecalsCount](#).

Example

[Add Decal \(C#\)](#)
[Add Decal \(VB.NET\)](#)
[Add Decal \(VBA\)](#)

Remarks

Call this method before calling [IFace2::IGetAllDecalProperties](#) to get the size of the array for that method.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::IGetAllDecalProperties Method](#)
[IFace2::IGetDecalProperties Method](#)
[IFace2::IGetDecalProperties Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetEdgeCount Method (IFace2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetEdgeCount Method (IFace2)

Gets the number of the edges that bound this face.

.NET Syntax

Visual Basic (Declaration)

```
Function GetEdgeCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As System.Integer  
  
value = instance.GetEdgeCount()
```

C#

```
System.int GetEdgeCount()
```

C++/CLI

```
System.int GetEdgeCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of edges

Visual Basic for Applications (VBA) Syntax

See [Face2::GetEdgeCount](#).

Example

[Traverse Bodies \(C++\)](#)
[Get Faces Affected by Feature \(VBA\)](#)
[Get Faces Affected by Feature \(VB.NET\)](#)
[Get Faces Affected by Feature \(C#\)](#)

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IEdge Interface](#)
[IFace2::GetEdges Method](#)
[IFace2::IGetEdges Method](#)
[IFace2::EnumEdges Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetEdges Method (IFace2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetEdges Method (IFace2)

Get the edges bounding this face.

.NET Syntax

Visual Basic (Declaration)

```
Function GetEdges() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As System.Object  
  
value = instance.GetEdges()
```

C#

```
System.object GetEdges()
```

C++/CLI

```
System.Object^ GetEdges();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [edges](#)

Visual Basic for Applications (VBA) Syntax

See [Face2::GetEdges](#).

Example

[Add Highlighting To or Remove Highlighting From Edges \(VBA\)](#)
[Check Interference Using IModeler::CheckInterference \(VBA\)](#)
[Get Faces Affected by Feature \(VBA\)](#)
[Get Vertex \(VBA\)](#)

Remarks

If you want to align the edges with their corresponding trim curves, then call this method immediately following a call to [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#). This causes the order of the trim curves to align with the order of the edges. For Dispatch implementations, use [IFace2::GetEdges](#).

To elaborate, any call to [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#) stores a temporary buffer containing the edges for this face. If this temporary buffer exists, then [IFace2::GetEdges](#), [IFace2::IGetEdges](#), or [IFace2::EnumEdges](#) access the buffer. This is useful when you want to align the trim curves of the face with their corresponding edges. If [IFace2::GetEdges](#), [IFace2::IGetEdges](#), or [IFace2::EnumEdges](#) is called immediately after [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#), then the order of the trim curves align with the order of the edges. However, if you are processing all the trim curves and then processing all the edges, your first call to [IFace2::EnumEdges](#) uses the temporary buffer of edges generated by your last call to [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#). Because your last call to [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#) and your current call to [IFace2::EnumEdges](#) are probably made from different faces, the data does not match. You can see this mismatch of data when [IFace2::GetEdgeCount](#) does not match the number of enumerated edges, or when the edge data does not match the examined face. Therefore, call [IFace2::GetEdges](#), [IFace2::IGetEdges](#), or [IFace2::EnumEdges](#) in the same loop and immediately after your call to [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#). The temporary buffer of edges is destroyed after each call to [IFace2::GetEdges](#), [IFace2::IGetEdges](#), or [IFace2::EnumEdges](#).

If you are using [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#), the number of trim curves on a face might vary from the number of edges on the face. If you call [IFace2::EnumEdges](#) immediately after calling [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#), then you are getting list of edges generated by [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#). Because this list of generated edges might vary from the list of edges that are typically returned from this face, do not use [IFace2::GetEdgeCount](#) to determine the length of the enumerated edge list. Instead, use one of the return values from [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#) to determine the size of the enumerated edge list. [IFace2::GetTrimCurves2](#) or [IFace2::IGetTrimCurves2](#) returns the number of trim (SP) curves in the first array position. This is the value you can use to loop through your enumerated list of edges.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::EnumEdges Method](#)
[IFace2::GetEdgeCount Method](#)
[IFace2::IGetEdges Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetFaceId Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : [GetFaceId Method \(IFace2\)](#)

Gets the face ID of an imported body.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFaceId() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As System.Integer  
  
value = instance.GetFaceId()
```

C#

```
System.int GetFaceId()
```

C++/CLI

```
System.int GetFaceId();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Face ID

Visual Basic for Applications (VBA) Syntax

See [Face2::GetFacelD](#).

Remarks

SOLIDWORKS uses face IDs to track specific faces of imported bodies (for example, IGES imports).

Prior to [setting a face ID](#), you should examine all faces in the model to get their face IDs, if any, to ensure that the face ID you are setting is unique to the model.

Face IDs are saved with the document, but face IDs are removed whenever the document is rebuilt. Typically you assign face IDs to a read-only copy of the finalized model.

Any third-party application can change a face ID. The intent is that you assign face IDs so that you can refer to those faces within your application.

Use the [IAttribute](#) object to store data with a face.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::RemoveFacelD Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetFeature Method (IFace2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetFeature Method (IFace2)

Gets the feature that owns this face.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFeature() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFace2
Dim value As System.Object

value = instance.GetFeature()
```

C#

```
System.object GetFeature()
```

C++/CLI

```
System.Object^ GetFeature();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Owning [feature](#)

Visual Basic for Applications (VBA) Syntax

See [Face2::GetFeature](#).

Example

[Get Part for Corresponding Component \(VBA\)](#)
[Get Faces Associated with Feature \(C#\)](#)
[Get Faces Associated with Feature \(VB.NET\)](#)
[Get Faces Associated with Feature \(VBA\)](#)

Remarks

In SOLIDWORKS, a face:

- is the result of evaluating a feature.
- can be owned by several features.

[IFeature::GetFaces](#) or [IFeature::IGetFaces2](#) returns all of the faces owned by a feature. This is different from the faces highlighted in the user interface when the feature is selected. The user interface filters out multiple feature faces. This filter is only for display purposes.

To filter out multiple feature faces using the SOLIDWORKS API, you must call [IFace2::GetFeature](#) or [IFace2::IGetFeature](#). Only the oldest feature from the face is returned, that is, the first owning feature in the FeatureManager design tree.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetFeatureId Method](#)
[IFace2::IGetFeature Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetFeatureId Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetFeatureId Method (IFace2)

Gets the order number for the owning feature of this face.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFeatureId() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As System.Integer  
  
value = instance.GetFeatureId()
```

C#

```
System.int GetFeatureId()
```

C++/CLI

```
System.int GetFeatureId();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Order number of the owning feature

Visual Basic for Applications (VBA) Syntax

See [Face2::GetFeatureId](#).

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetFeature Method](#)
[IFace2::IGetFeature Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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GetFirstLoop Method (IFace2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetFirstLoop Method (IFace2)

Gets the first loop in this face, which is not necessarily the outer loop.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFirstLoop() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As System.Object  
  
value = instance.GetFirstLoop()
```

C#

```
System.object GetFirstLoop()
```

C++/CLI

```
System.Object^ GetFirstLoop();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

First [loop](#)

Visual Basic for Applications (VBA) Syntax

See [Face2::GetFirstLoop](#).

Example

[Determine Type of Face \(VBA\)](#)
[Find Outside Edges of Face \(VBA\)](#)
[Get Loops \(VBA\)](#)
[Select Tangent Faces \(VBA\)](#)
[Sweep Planar Loop Along Vector \(VBA\)](#)
[Fill Holes in Part \(C#\)](#)
[Fill Holes in Part \(VB.NET\)](#)
[Fill Holes in Part \(VBA\)](#)
[Select Edges of All Holes on Face \(C#\)](#)
[Select Edges of All Holes on Face \(VB.NET\)](#)
[Select Edges of All Holes on Face \(VBA\)](#)

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetFirstLoop Method](#)
[IFace2::GetLoopCount Method](#)
[ILoop2::GetNext Method](#)
[ILoop2::IGetNext Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetLoopCount Method (IFace2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetLoopCount Method (IFace2)

Gets the number of loops in this face.

.NET Syntax

Visual Basic (Declaration)

```
Function GetLoopCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As System.Integer  
  
value = instance.GetLoopCount()
```

C#

```
System.int GetLoopCount()
```

C++/CLI

```
System.int GetLoopCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of loops in this face

Visual Basic for Applications (VBA) Syntax

See [Face2::GetLoopCount](#).

Example

[Find Outside Edges of Face \(VBA\)](#)
[Get Loops \(VBA\)](#)

Remarks

Call this method before calling [IFace2::!GetLoops](#) to determine the size of the array for that method.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[ILoop2 Interface](#)
[IFace2::GetFirstLoop Method](#)
[IFace2::!GetFirstLoop Method](#)
[ILoop2::GetNext Method](#)
[ILoop2::!GetNext Method](#)
[IFace2::GetLoops Method](#)
[IFace2::!GetLoops Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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GetLoops Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetLoops Method (IFace2)

Gets all of the loops on this face.

.NET Syntax

Visual Basic (Declaration)

Function GetLoops() As System.Object

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As System.Object  
  
value = instance.GetLoops()
```

C#

```
System.object GetLoops()
```

C++/CLI

```
System.Object^ GetLoops();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of all of the [loops](#) on the face (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Face2::GetLoops](#).

Remarks

If a face has one or more holes on it, then this method gets the one loop for the face and a loop for every hole on the face.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::IGetLoops Method](#)
[IFace2::IGetFirstLoop Method](#)
[IFace2::GetFirstLoop Method](#)
[IFace2::GetLoopCount Method](#)
[ILoop2::GetNext Method](#)
[ILoop2::IGetNext Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetMaterialPropertyValues2 Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetMaterialPropertyValues2 Method (IFace2)

Gets the material property values for this face.

.NET Syntax

Visual Basic (Declaration)

```
Function GetMaterialPropertyValues2( _  
    ByVal Config\_opt As System.Integer, _  
    ByVal Config\_names As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim Config_opt As System.Integer  
Dim Config_names As System.Object  
Dim value As System.Object
```

```
value = instance.GetMaterialPropertyValues2(Config_opt, Config_names)
```

C#

```
System.object GetMaterialPropertyValues2(  
    System.int Config\_opt,  
    System.object Config\_names  
)
```

C++/CLI

```
System.Object^ GetMaterialPropertyValues2(  
& System.int Config\_opt,  
& System.Object^ Config\_names  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Config_opt

Configuration option as defined in [swInConfigurationOpts_e](#)

Config_names

Array of configuration names (see **Remarks**)

Return Value

Array of material property values of the material for this face (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Face2::GetMaterialPropertyValues2](#).

Remarks

The material property values returned include the color (R,G,B values), reflectivity (ambient, diffuse, specular, shininess), transparency, and emission.

If you set Config_opt to swThisConfiguration or swAllConfiguration, then pass Nothing because Config_names is ignored.

The format of material_values is an array of doubles:

[*R, G, B, Ambient, Diffuse, Specular, Shininess, Transparency, Emission*]

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::IGetMaterialPropertyValues2 Method](#)
[IFace2::IRemoveMaterialProperty2 Method](#)
[IFace2::ISetMaterialPropertyValues2 Method](#)
[IFace2::RemoveMaterialProperty2 Method](#)
[IFace2::SetMaterialPropertyValues2 Method](#)
[IFace2::IMaterialPropertyValues Property](#)
[IFace2::MaterialPropertyValues Property](#)
[IFeature::GetMaterialPropertyValues2 Method](#)
[IFeature::IGetMaterialPropertyValues2 Method](#)
[IModelDoc2::IMaterialPropertyValues Property](#)
[IModelDoc2::MaterialPropertyValues Property](#)
[IModelDocExtension::GetMaterialPropertyValues Method](#)
[IModelDocExtension::IGetMaterialPropertyValues Method](#)
[IPartDoc::IMaterialPropertyValues Property](#)
[IPartDoc::MaterialPropertyValues Property](#)
[IFace2::HasMaterialPropertyValues Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetNextFace Method (IFace2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetNextFace Method (IFace2)

Gets the next face in a body.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function GetNextFace() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As System.Object  
  
value = instance.GetNextFace()
```

C#

```
System.Object GetNextFace()
```

C++/CLI

```
System.Object^ GetNextFace();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Next [face](#) in a body

Visual Basic for Applications (VBA) Syntax

See [Face2::GetNextFace](#).

Example

[Get Part Bounding Box \(VBA\)](#)
[Get Material Properties \(VBA\)](#)
[Get Part Bounding Box \(VBA\)](#)
[Traverse Bodies \(C++\)](#)
[Set Body for View \(C#\)](#)
[Set Body for View \(VB.NET\)](#)
[Set Body for View \(VBA\)](#)
[Delete Blended Faces \(C#\)](#)
[Delete Blended Faces \(VB.NET\)](#)
[Delete Blended Faces \(VBA\)](#)

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IBody2::GetFirstFace Method](#)
[IBody2::IGetFirstFace Method](#)
[IBody2::IGetFirstSelectedFace Method](#)
[IBody2::GetFirstSelectedFace Method](#)
[IFace2::IGetNextFace Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetPatternSeedFeature Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetPatternSeedFeature Method (IFace2)

Gets the seed feature of a pattern.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function GetPatternSeedFeature() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As System.Object  
  
value = instance.GetPatternSeedFeature()
```

C#

```
System.Object GetPatternSeedFeature()
```

C++/CLI

```
System.Object^ GetPatternSeedFeature();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Seed feature of the current face; if the face does not belong to a pattern, this method returns NULL

Visual Basic for Applications (VBA) Syntax

See [Face2::GetPatternSeedFeature](#).

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::IGetPatternSeedFeature Method](#)
[IModelDoc2::EditSeedFeat Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetProjectedPointOn Method (IFace2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetProjectedPointOn Method (IFace2)

Gets the point where the input point is projected on to this face.

.NET Syntax

Visual Basic (Declaration)

```
Function GetProjectedPointOn( _  
    ByVal Point As MathPoint, _  
    ByVal Direction As MathVector _  
) As MathPoint
```

Visual Basic (Usage)

```
Dim instance As IFace2  
Dim Point As MathPoint  
Dim Direction As MathVector  
Dim value As MathPoint
```

```
value = instance.GetProjectedPointOn(Point, Direction)
```

C#

```
MathPoint GetProjectedPointOn(  
    MathPoint Point,  
    MathVector Direction  
)
```

C++/CLI



```
MathPoint^ GetProjectedPointOn(
& MathPoint^ Point,
& MathVector^ Direction
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Point
[Point](#) to project
Direction
[Direction](#) to project the point

Return Value

[Point](#) where the input point is projected on to the face

Visual Basic for Applications (VBA) Syntax

See [Face2::GetProjectedPointOn](#).

Example

[Get Projected Point \(VBA\)](#)

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetClosestPointOn Method](#)
[IFace2::IGetClosestPointOn Method](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetPropertyExtension Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetPropertyExtension Method (IFace2)

Gets the property extension on this face.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPropertyExtension( _
    ByVal ID As System.Integer _
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As IFace2
Dim ID As System.Integer
Dim value As System.Object

value = instance.GetPropertyExtension(ID)
```

C#

```
System.object GetPropertyExtension(
    System.int ID
)
```

C++/CLI



```
System.Object^ GetPropertyExtension(  
& System.int ID  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ID
0

Return Value

Value of the property extension

Visual Basic for Applications (VBA) Syntax

See [Face2::GetPropertyExtension](#).

Remarks

SOLIDWORKS recommends that you use the [IAttribute](#), [IAttributeDef](#), and [IParameter](#) interfaces instead of this method. These interfaces provide more flexibility.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::ResetPropertyExtension Method](#)
[IFace2::AddPropertyExtension Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetSeedFeature Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetSeedFeature Method (IFace2)

Gets the seed feature of a patterned, mirrored, or copied body for this face.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSeedFeature() As Feature
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As Feature
```

```
value = instance.GetSeedFeature()
```

C#

```
Feature GetSeedFeature()
```

C++/CLI

```
Feature^ GetSeedFeature();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Seed [feature](#) of a patterned, mirrored, or copied body

Visual Basic for Applications (VBA) Syntax

See [Face2::GetSeedFeature](#).

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetPatternSeedFeature Method](#)

Availability

SOLIDWORKS 2008 SP4, Revision Number 16.4

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetShellType Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetShellType Method (IFace2)

Gets the shell type for this face.

.NET Syntax

Visual Basic (Declaration)

```
Function GetShellType() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As System.Integer  
  
value = instance.GetShellType()
```

C#

```
System.int GetShellType()
```

C++/CLI

```
System.int GetShellType();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Shell type:

- 0 = Open shell. For example, a face that belongs to a sheet (surface) body or reference surface.
- 1 = Internal shell. For example, a face that belongs to a cavity. Face helps define an internal volume.
- 2 = External shell. For example, a typical face on a solid body (helps "hold in" the body mass). This includes all external faces (faces belonging to bosses, pockets, holes, and so on).

Visual Basic for Applications (VBA) Syntax

See [Face2::GetShellType](#).

See Also

[IFace2 Interface](#)
[IFace2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetSilhoutteEdges Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetSilhoutteEdges Method (IFace2)

Obsolete. Superseded by [IFace2::GetSilhoutteEdgesVB](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetSilhoutteEdges( _  
    ByRef Root As System.Double, _  
    ByRef Normal As System.Double _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim Root As System.Double  
Dim Normal As System.Double  
Dim value As System.Object  
  
value = instance.GetSilhoutteEdges(Root, Normal)
```

C#

```
System.object GetSilhoutteEdges(  
    ref System.double Root,  
    ref System.double Normal  
)
```

C++/CLI

```
System.Object^ GetSilhoutteEdges(  
& System.double% Root,  
& System.double% Normal  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Root

Normal

Visual Basic for Applications (VBA) Syntax

See [Face2::GetSilhoutteEdges](#).

See Also

[IFace2 Interface](#)
[IFace2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetSilhoutteEdgesVB Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetSilhoutteEdgesVB Method (IFace2)

Gets the silhouette edges.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSilhoutteEdgesVB( _  
    ByVal Xroot As System.Double, _
```

```

    ByVal Yroot As System.Double, _
    ByVal Zroot As System.Double, _
    ByVal Xnormal As System.Double, _
    ByVal Ynormal As System.Double, _
    ByVal Znormal As System.Double _
) As System.Object

```

Visual Basic (Usage)



```

Dim instance As IFace2
Dim Xroot As System.Double
Dim Yroot As System.Double
Dim Zroot As System.Double
Dim Xnormal As System.Double
Dim Ynormal As System.Double
Dim Znormal As System.Double
Dim value As System.Object

value = instance.GetSilhoutteEdgesVB(Xroot, Yroot, Zroot, Xnormal, Ynormal, Znormal)

```

C#

```

System.object GetSilhoutteEdgesVB(
    System.double Xroot,
    System.double Yroot,
    System.double Zroot,
    System.double Xnormal,
    System.double Ynormal,
    System.double Znormal
)

```

C++/CLI

```

System.Object^ GetSilhoutteEdgesVB(
    & System.double Xroot,
    & System.double Yroot,
    & System.double Zroot,
    & System.double Xnormal,
    & System.double Ynormal,
    & System.double Znormal
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Xroot

X component of the root point

Yroot

Y component of the root point

Zroot

Z component of the root point

Xnormal

X component of the direction vector

Ynormal

Y component of the direction vector

Znormal

Z component of the direction vector

Return Value

Array of [edges](#)

Visual Basic for Applications (VBA) Syntax

See [Face2::GetSilhoutteEdgesVB](#).

Remarks

The return array has two elements for each edge: the first is the silhouette edge and the second is unused. To iterate through the edges, an application needs to step through every second element.

The returned edges are transient and cannot be selected.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::IGetSilhouetteEdges Method](#)
[IFace2::IGetSilhouetteEdgeCount Method](#)
[IModelDoc2::InsertSplitLineSil Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetSurface Method (IFace2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetSurface Method (IFace2)

Gets the surface referenced by this face.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSurface() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As System.Object  
  
value = instance.GetSurface()
```

C#

```
System.object GetSurface()
```

C++/CLI

```
System.Object^ GetSurface();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Underlying [surface](#) for this face

Visual Basic for Applications (VBA) Syntax

See [Face2::GetSurface](#).

Example

[Calculate Closest Distance Between Faces \(VBA\)](#)
[Determine Type of Face \(VBA\)](#)
[Get Intersecting Face and Edge \(VBA\)](#)
[Get Intersecting Faces \(VBA\)](#)
[Get Parameters of Planar Surface \(VBA\)](#)
[Get Parameters of Toroidal Surface \(VBA\)](#)
[Get Surface Type \(VBA\)](#)
[Select Tangent Faces \(VBA\)](#)
[Fill Holes in Part \(C#\)](#)
[Fill Holes in Part \(VB.NET\)](#)
[Fill Holes in Part \(VBA\)](#)
[Select Edges of All Holes on Face \(C#\)](#)
[Select Edges of All Holes on Face \(VB.NET\)](#)
[Select Edges of All Holes on Face \(VBA\)](#)

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::IGetSurface Method](#)
[IFace2::AttachSurface Method](#)
[IFace2::DetachSurface Method](#)
[IFace2::HasTextureCoordinates Method \(\)](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetTessNorms Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetTessNorms Method (IFace2)

Gets the normal vector for each of the triangles that make up the shaded picture tessellation.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTessNorms() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As System.Object  
  
value = instance.GetTessNorms()
```

C#

```
System.object GetTessNorms()
```

C++/CLI

```
System.Object^ GetTessNorms();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Face2::GetTessNorms](#).

Remarks

The data returned by this method takes the following format:

- float x, y, z - First point's unit normal
- float x, y, z - Second point's unit normal
- float x, y, z - Third point's unit normal

where this data repeats itself for the set of triangles on this IFace2.

The total size of the data is [9 x sizeof(float) x (number of triangles)].

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::IGetTessNorms Method](#)
[IFace2::GetTessTextures Method](#)
[IFace2::GetTessTriangleCount Method](#)
[IFace2::GetTessTriangles Method](#)
[IFace2::GetTessTriStripEdges Method](#)
[IFace2::GetTessTriStripNorms Method](#)
[IFace2::GetTessTriStrips Method](#)

[IFace2::GetTessTriStripSize Method](#)
[IFace2::GetTessTextures Method](#)
[IFace2::GetTessTriangles Method](#)
[IFace2::GetTessTriStripEdges Method](#)
[IFace2::GetTessTriStripEdgeSize Method](#)
[IFace2::GetTessTriStripNorms Method](#)
[IFace2::GetTessTriStrips Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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GetTessTextures Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetTessTextures Method (IFace2)

Gets the texture coordinates for the triangles.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTessTextures() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As System.Object  
  
value = instance.GetTessTextures()
```

C#

```
System.object GetTessTextures()
```

C++/CLI

```
System.Object^ GetTessTextures();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of doubles containing the texture coordinates for the triangles

Visual Basic for Applications (VBA) Syntax

See [Face2::GetTessTextures](#).

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetTessTextures Method](#)
[IFace2::GetTessNorms Method](#)
[IFace2::GetTessTriangleCount Method](#)
[IFace2::GetTessTriangles Method](#)
[IFace2::GetTessTriStripEdges Method](#)
[IFace2::GetTessTriStripNorms Method](#)
[IFace2::GetTessTriStrips Method](#)
[IFace2::GetTessTriStripSize Method](#)
[IFace2::GetTessNorms Method](#)
[IFace2::GetTessTriangles Method](#)
[IFace2::GetTessTriStripEdges Method](#)
[IFace2::GetTessTriStripEdgeSize Method](#)
[IFace2::GetTessTriStripNorms Method](#)
[IFace2::GetTessTriStrips Method](#)
[IPartDoc::GetTessTriStripTextures Method \(\)](#)
[IFace2::HasTextureCoordinates Method \(\)](#)

Availability

SOLIDWORKS 2005 SP2, Revision Number 13.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetTessTriangleCount Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetTessTriangleCount Method (IFace2)

Gets the number of triangles that make up the shaded picture tessellation for this face.

.NET Syntax

Visual Basic (Declaration)

Function GetTessTriangleCount() As System.Integer

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As System.Integer  
  
value = instance.GetTessTriangleCount()
```

C#

System.int GetTessTriangleCount()

C++/CLI

System.int GetTessTriangleCount();

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Face2::GetTessTriangleCount](#).

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetTessNorms Method](#)
[IFace2::GetTessTextures Method](#)
[IFace2::GetTessTriangles Method](#)
[IFace2::GetTessTriStripEdges Method](#)
[IFace2::GetTessTriStripNorms Method](#)
[IFace2::GetTessTriStripSize Method](#)
[IFace2::GetTessTriStripSize Method](#)
[IFace2::IGetTessNorms Method](#)
[IFace2::IGetTessTextures Method](#)
[IFace2::IGetTessTriangles Method](#)
[IFace2::IGetTessTriStripEdges Method](#)
[IFace2::IGetTessTriStripEdgeSize Method](#)
[IFace2::IGetTessTriStripNorms Method](#)
[IFace2::IGetTessTriStripSize Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetTessTriangles Method (IFace2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetTessTriangles Method (IFace2)

Gets the triangles that make up the shaded picture tessellation for this face.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTessTriangles( _  
    ByVal NoConversion As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As IFace2
Dim NoConversion As System.Boolean
Dim value As System.Object

value = instance.GetTessTriangles(NoConversion)
```

C#

```
System.Object GetTessTriangles(
    System.bool NoConversion
)
```

C++/CLI

```
System.Object^ GetTessTriangles(
    & System.bool NoConversion
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NoConversion

True prohibits conversion to user units from system units, false does not

Return Value

Array (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Face2::GetTessTriangles](#).

Example

[Get Part Bounding Box \(VBA\)](#)

Remarks

These triangles are intended for graphics display purposes and do not represent a tessellation that can be used, for example, by a machining application. If you need the kind of accuracy associated with a machining product, traverse the body faces and extract the topology and geometry data to create your own faceting.

The format of the returned data is:

float x, y, z - First point in meters

float x, y, z - Second point in meters

float x, y, z - Third point in meters

where this data repeats itself for each of the triangles on this face.

The total size of the data is $\left[9 \times \text{sizeof}(\text{float}) \times (\text{number of triangles}) \right]$.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetTessNorms Method](#)
[IFace2::GetTessTextures Method](#)
[IFace2::GetTessTriangleCount Method](#)
[IFace2::GetTessTriStripEdges Method](#)
[IFace2::GetTessTriStripNorms Method](#)
[IFace2::GetTessTriStrips Method](#)
[IFace2::GetTessTriStripSize Method](#)
[IFace2::IGetTessNorms Method](#)
[IFace2::IGetTessTextures Method](#)
[IFace2::IGetTessTriangles Method](#)
[IFace2::IGetTessTriStripEdges Method](#)
[IFace2::IGetTessTriStripEdgeSize Method](#)
[IFace2::IGetTessTriStripNorms Method](#)
[IFace2::IGetTessTriStrips Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetTessTriStripEdges Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetTessTriStripEdges Method (IFace2)

Gets the edge ID for each of the edges in the triangles that make up the tessellation for this face.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTessTriStripEdges() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As System.Object
```

```
value = instance.GetTessTriStripEdges()
```

C#

```
System.Object GetTessTriStripEdges()
```

C++/CLI

```
System.Object^ GetTessTriStripEdges();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array that contains the list of edge IDs for this face (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Face2::GetTessTriStripEdges](#).

Remarks

Returns an array of long or integers (see [Long vs. Integer](#)) values that indicate which edges of the given triangle strip represent a face boundary. The array takes the form of:

```
[nStrips,  
 stripEdgeCount1, stripEdgeCount2,..., stripVertexCountN,  
 strip1[triStripFlag, stripEdgeCount1 elements],  
 strip2[triStripFlag, stripEdgeCount2 elements],...,  
 stripN[triStripFlag, stripEdgeCountN elements]&cd;
```

where:

nStrips = The number of triangle strips on the face.

stripEdgeCountN = The number of triangle edges for the *N*th triangle strip + 1.

stripN = A sub-array that consists of the *triStripFlag* and an array of *stripEdgeCountN* *edgeIds*.

triStripFlag = Indicates if the triangle strip is a strip (=1) or if the triangle strip is a triangle (=0).

edgeId = An identifier with an arbitrary value that represents the edge of the face. Each edge has a unique *edgeId*.

```
2---4---6---8---10  
 / \ / \ / \ / \ /  
 / \ / \ / \ / \ /  
 / \ / \ / \ / \ /  
 / \ / \ / \ / \ /  
 / \ / \ / \ / \ /  
1---3---5---7---9
```

If you have this triangle strip, then you get a *stripVertexCount* of 10 using [IFace2::GetTessTriStrips](#) or [IFace2::IGetTessTriStrips](#). [IFace2::GetTessTriStripEdges](#) and [IFace2::IGetTessTriStripEdges](#) get a *stripEdgeCount* of 18. This means that for each triangle strip $stripEdgeCount - 1 = 2(stripVertexCount - 2) + 1$. The first element represents a flag indicating if the current strip is a triangle strip or a simple triangle, 1 or 0, respectively.

If one of the triangle strip edges lies on a face edge, then the value of $stripN[i] = edgeId$. If the triangle strip edge does not lie on a face edge, then the value of $stripN[i] = 0$.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::GetTessNorms Method](#)
[IFace2::GetTessTextures Method](#)
[IFace2::GetTessTriangleCount Method](#)
[IFace2::GetTessTriangles Method](#)
[IFace2::GetTessTriStripNorms Method](#)
[IFace2::GetTessTriStrips Method](#)
[IFace2::GetTessTriStripSize Method](#)
[IFace2::IGetTessNorms Method](#)
[IFace2::IGetTessTextures Method](#)
[IFace2::IGetTessTriangles Method](#)
[IFace2::IGetTessTriStripEdges Method](#)
[IFace2::IGetTessTriStripEdgeSize Method](#)
[IFace2::IGetTessTriStripNorms Method](#)
[IFace2::IGetTessTriStrips Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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GetTessTriStripNorms Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetTessTriStripNorms Method (IFace2)

Gets the normal vector for each of the triangles that make up the shaded picture tessellation for this face.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTessTriStripNorms() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim value As System.Object  
  
value = instance.GetTessTriStripNorms()
```

C#

```
System.object GetTessTriStripNorms()
```

C++/CLI

```
System.Object^ GetTessTriStripNorms();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array that contains the tristrip normals (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Face2::GetTessTriStripNorms](#).

Remarks

The format of the returned data is:

DWORD *NumStrips*

DWORD [*VertexPerStrip*]

Float [*Normals*]

where:

- NumStrips* = Number of strips on a particular face
- VertexPerStrip* = Array containing the number of vertex points on each face strip

- *Normals* = Array of X,Y,Z normal components for each vertex on each face strip

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::IGetTessTriStripNormals Method](#)
[IFace2::GetTessNormals Method](#)
[IFace2::GetTessTextures Method](#)
[IFace2::GetTessTriangleCount Method](#)
[IFace2::GetTessTriangles Method](#)
[IFace2::GetTessTriStripEdges Method](#)
[IFace2::GetTessTriStrips Method](#)
[IFace2::GetTessTriStripSize Method](#)
[IFace2::IGetTessNormals Method](#)
[IFace2::IGetTessTextures Method](#)
[IFace2::IGetTessTriangles Method](#)
[IFace2::IGetTessTriStripEdges Method](#)
[IFace2::IGetTessTriStripEdgeSize Method](#)
[IFace2::IGetTessTriStrips Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetTessTriStrips Method (IFace2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFace2 Interface](#) : GetTessTriStrips Method (IFace2)

Gets the vertices that make up the shaded picture tessellation for this face.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTessTriStrips( _  
    ByVal NoConversion As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFace2  
Dim NoConversion As System.Boolean  
Dim value As System.Object  
  
value = instance.GetTessTriStrips(NoConversion)
```

C#

```
System.object GetTessTriStrips(  
    System.bool NoConversion  
)
```

C++/CLI

```
System.Object^ GetTessTriStrips(  
& System.bool NoConversion  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NoConversion

True prohibits conversion to user units from system units, false does not

Return Value

Array (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Face2::GetTessTriStrips](#).

Remarks

These triangles are intended for graphics display purposes and do not represent a tessellation that can be used, for example, by a machining application. If you need the level of accuracy associated with a machining product, traverse the body faces and extract the topology and geometry data to create your own faceting.

This method returns an array of float values in the form of:

```
[NumStrips,  
    VertexPerStrip1, VertexPerStrip2,..., VertexPerStripN,  
    VertexPoints1[VertexPerStrip1 elements]*3,  
    VertexPoints2[VertexPerStrip2 elements]*3,...,  
    VertexPointsN[VertexPerStripN elements]*3]
```

where:

NumStrips = number of triangle strips on a particular face.

VertexPerStripN = number of vertex points for the *N*th triangle strip.

VertexPointsN = a sub-array of vertices consisting of three float values representing the x,y,z coordinate of the vertex; the sub-array's length is defined by the *VertexPerStripN* element in the array.

Because the values *NumStrips* and the elements of *VertexPerStrip* are long values packed into single values, you must [unpack](#) them before using them.

See Also

[IFace2 Interface](#)
[IFace2 Members](#)
[IFace2::IGetTessTriStrips Method](#)
[IFace2::IGetTessNorms Method](#)
[IFace2::IGetTessTextures Method](#)
[IFace2::IGetTessTriangleCount Method](#)
[IFace2::IGetTessTriangles Method](#)
[IFace2::IGetTessTriStripEdges Method](#)
[IFace2::IGetTessTriStripNorms Method](#)
[IFace2::IGetTessTriStripSize Method](#)
[IFace2::IGetTessNorms Method](#)
[IFace2::IGetTessTextures Method](#)
[IFace2::IGetTessTriangles Method](#)
[IFace2::IGetTessTriStripEdges Method](#)
[IFace2::IGetTessTriStripEdgeSize Method](#)
[IFace2::IGetTessTriStripNorms Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0